# STAT 516: Probability Review and Undirected Graphical Models

Liang-Cheng Chen

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**Reading guide**

English transcription of `notes/stat516-note1.pdf`, with corrected formulas and supplementary explanations. Source-page labels refer to the original PDF. Yellow boxes preserve handwritten TODOs; red boxes identify corrections; blue boxes add explanations. Reference: Yen-Chi Chen, *Lecture 1: Review on Probability and Statistics* (L1) and *Lecture 2: Graphical Models* (L2), supplied in `lecture-notes/`. Source page 15 is blank.

## 1 Common random variables (October 1, 2026)

*Handwritten source: 1–3; L1, Sections 1.2.1–1.2.2.*

### 1.1 Discrete distributions

**Bernoulli.** For $0 \le \theta \le 1$,

$$X \sim \mathrm{Ber}(\theta), \qquad \mathbb{P}(X = 1) = \theta, \quad \mathbb{P}(X = 0) = 1 - \theta. \tag{1}$$

For an event $A$, its indicator is a Bernoulli random variable:

$$\mathbf{1}_A(\omega) = \begin{cases} 1, & \omega \in A, \\ 0, & \omega \notin A, \end{cases} \qquad \mathbf{1}_A \sim \mathrm{Ber}(\mathbb{P}(A)). \tag{2}$$

**Binomial.** For a nonnegative integer $n$,

$$X \sim \mathrm{Bin}(n, \theta), \qquad \mathbb{P}(X = x) = \binom{n}{x} \theta^x (1 - \theta)^{n-x}, \quad x \in \{0, \dots, n\}. \tag{3}$$

The mass is zero elsewhere. If $Z_1, \dots, Z_n$ are independent and identically distributed (IID) $\mathrm{Ber}(\theta)$ variables, then $X = \sum_{i=1}^{n} Z_i$ has this distribution.

**Poisson.** This distribution models counts. For $\lambda > 0$,

$$X \sim \text{Poisson}(\lambda), \qquad \mathbb{P}(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}, \quad x \in \{0, 1, ...\}. \tag{4}$$

**Discrete uniform.** For a positive integer $k$,

$$X \sim \text{Unif}\{1, \dots, k\}, \qquad \mathbb{P}(X = x) = \frac{1}{k}, \quad x \in \{1, \dots, k\}. \tag{5}$$

**Geometric.** Run independent Bernoulli trials with success probability $0 < \theta \leq 1$. The number of trials $T$ through the first success satisfies

$$\mathbb{P}(T = t) = \theta(1 - \theta)^{t-1}, \qquad t = 1, 2, \dots. \tag{6}$$

**Correction: Trials versus failures (source p. 2)**

The handwritten description counts trials, but its support and formula count failures. L1, Section 1.2.1 uses the trials convention above. If $F = T - 1$ counts failures before the first success, the handwritten formula becomes correct:

$$\mathbb{P}(F = f) = \theta(1 - \theta)^f, \qquad f = 0, 1, \dots. \tag{7}$$

## 1.2 Continuous distributions

All densities below are zero outside the stated support.

**Normal.** For $\mu \in \mathbb{R}$ and $\sigma^2 > 0$,

$$X \sim N(\mu, \sigma^2), \qquad p_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}. \tag{8}$$

**Uniform.** For $a < b$,

$$X \sim \text{Unif}[a, b], \qquad p_X(x) = \frac{1}{b - a} \mathbf{1}_{\{a \leq x \leq b\}}. \tag{9}$$

**Exponential.** With rate $\lambda > 0$,

$$X \sim \text{Exp}(\lambda), \qquad p_X(x) = \lambda e^{-\lambda x} \mathbf{1}_{\{x \geq 0\}}. \tag{10}$$

**Beta.** For $\alpha, \beta > 0$,

$$X \sim \mathrm{B}(\alpha, \beta), \qquad p_X(x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha,\beta)}, \quad 0 < x < 1, \tag{11}$$

$$B(\alpha,\beta) = \int_0^1 x^{\alpha-1}(1-x)^{\beta-1}\,dx = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}. \tag{12}$$

**Interpreting the center**

The handwritten "center" $\alpha/(\alpha+\beta)$ is the mean. It need not be the mode: when $\alpha, \beta > 1$, the mode is $(\alpha-1)/(\alpha+\beta-2)$; for other parameter values the density may be monotone or U-shaped.

**Gamma.** We use the shape–rate parameterization:

$$X \sim \mathrm{Gamma}(\alpha, \lambda), \qquad p_X(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}, \quad x > 0, \quad \alpha, \lambda > 0. \tag{13}$$

In particular, $\mathrm{Gamma}(1, \lambda) = \mathrm{Exp}(\lambda)$ in distribution.

**Chi-square.** If $Z_1, \ldots, Z_d$ are IID $N(0,1)$, then

$$X = \sum_{i=1}^{d} Z_i^2 = \|Z\|^2 \sim \chi^2_d = \mathrm{Gamma}\left(\frac{d}{2}, \frac{1}{2}\right) \quad \text{in distribution.} \tag{14}$$

**Correction: Chi-square parameters (source p. 3)**

The handwritten Gamma parameters are reversed. With the shape–rate convention used here and in L1, Section 1.2.2, the shape is $d/2$ and the rate is $1/2$.

## 1.3 CDF, PDF, and PMF

*Handwritten source: 3; L1, Sections 1.1.2 and 1.2.3.*

The cumulative distribution function (CDF) always characterizes a real-valued distribution:

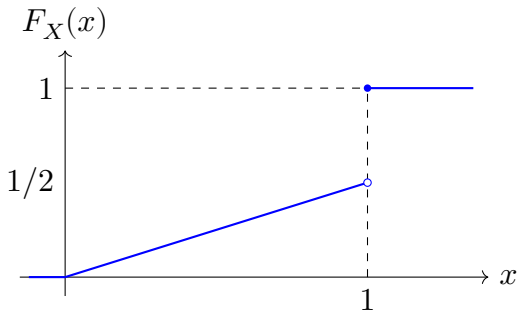
$$F_X(x) = \mathbb{P}(X \le x). \tag{15}$$

For an absolutely continuous distribution, $p_X(x) = F_X'(x)$ almost everywhere. At any point, the atom is

$$\mathbb{P}(X = x) = F_X(x) - F_X(x^-), \qquad F_X(x^-) = \lim_{\varepsilon \downarrow 0} F_X(x - \varepsilon). \tag{16}$$

**Mixed-distribution example.** With probability $1/2$ set $X = 1$; otherwise draw $X \sim \mathrm{Unif}(0, 1)$. Then

$$F_X(x) = \begin{cases} 0, & x < 0, \\ x/2, & 0 \le x < 1, \\ 1, & x \ge 1. \end{cases} \tag{17}$$




**Correction: The jump in the mixed CDF (source p. 3)**

The linear part reaches $1/2$, not 1, as $x \uparrow 1$; the CDF then jumps to 1. Its derivative captures only the uniform component, while the point masses capture only the atom at 1. Neither a Lebesgue PDF alone nor a discrete PMF alone describes the entire law. L1, Example 1.1 illustrates the same principle with a different mixture; its stated Bernoulli weight and atom probability contain an apparent mismatch, so they are not copied here.

# 2 Joint distributions and conditioning

*Handwritten source: 4–5 (October 6, 2026); L1, Section 1.3.1.*

## 2.1 Independence and joint densities

For arbitrary real-valued variables,

$$X \perp\!\!\!\perp Y \quad \Longleftrightarrow \quad F_{XY}(x, y) = F_X(x) F_Y(y) \quad \text{for all } x, y. \tag{18}$$

When a joint density exists relative to the relevant product reference measure, independence is equivalent to $p_{XY}(x, y) = p_X(x) p_Y(y)$ almost everywhere. This includes mixed discrete–continuous pairs.

**Measure-theoretic interpretation**

Independence is a central additional structure in probability theory. It is defined rigorously within measure theory; the handwritten comment should not be read as saying that measure theory cannot express independence. A density is a Radon–Nikodym derivative relative to a specified reference measure.

If $X$ is continuous and $Y$ is discrete, the joint density is with respect to Lebesgue measure in $x$ and counting measure in $y$:

$$p_{XY}(x,y) = p_{X|Y}(x \mid y)p_Y(y) = p_{Y|X}(y \mid x)p_X(x), \tag{19}$$

$$p_X(x) = \sum_y p_{XY}(x,y), \qquad p_Y(y) = \int_{\mathbb{R}} p_{XY}(x,y)\,dx. \tag{20}$$

Conditional-density ratios are used only where their denominators are positive, and conditional laws are defined up to marginal null sets.

## 2.2 Beta–Bernoulli example

*Handwritten source: 5–7; compare L1, Section 1.3.1.*

Suppose

$$X \sim \mathrm{B}(\alpha,\beta), \qquad Y \mid X = x \sim \mathrm{Ber}(x). \tag{21}$$

The joint density and the marginal of $X$ are

$$p_{XY}(x,y) = \frac{x^{\alpha+y-1}(1-x)^{\beta-y}}{B(\alpha,\beta)}, \quad 0<x<1, \quad y \in \{0,1\}, \tag{22}$$

$$\sum_{y=0}^{1} p_{XY}(x,y) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha,\beta)}\{x+(1-x)\} = p_X(x). \tag{23}$$

For fixed $y$, regard $y$ as constant and normalize over $x$:

$$p_Y(y) = \frac{B(\alpha+y,\beta+1-y)}{B(\alpha,\beta)}, \tag{24}$$

$$p_{X|Y}(x \mid y) = \frac{x^{\alpha+y-1}(1-x)^{\beta+1-y-1}}{B(\alpha+y,\beta+1-y)}, \tag{25}$$

$$X \mid Y = y \sim \mathrm{B}(\alpha+y,\beta+1-y). \tag{26}$$

The prior and posterior belong to the same family: the Beta family is conjugate to a Bernoulli likelihood. The marginal distribution of $Y$ can be obtained in two ways:

$$\mathbb{P}(Y=1) = \int_0^1 x\,p_X(x)\,dx = \frac{\alpha}{\alpha+\beta}, \tag{27}$$

$$\mathbb{E}[Y] = \mathbb{E}\{\mathbb{E}[Y \mid X]\} = \mathbb{E}[X] = \frac{\alpha}{\alpha+\beta}. \tag{28}$$

Since $Y$ is binary, either calculation identifies $Y \sim \mathrm{Ber}(\alpha/(\alpha+\beta))$.

## 2.3 Law of total variance

*Handwritten source: 8; L1, Section 1.3.3.*

Assume $\mathbb{E}[Y^2] < \infty$ and write $g(X) = \mathbb{E}[Y \mid X]$. Then

$$\begin{aligned} \operatorname{Var}(Y) &= \mathbb{E}[Y^2] - (\mathbb{E}[Y])^2 && (29)\\ &= \mathbb{E}\{\mathbb{E}[Y^2 \mid X]\} - \mathbb{E}[g(X)^2] + \mathbb{E}[g(X)^2] - (\mathbb{E}[g(X)])^2 && (30)\\ &= \mathbb{E}\{\operatorname{Var}(Y \mid X)\} + \operatorname{Var}(\mathbb{E}[Y \mid X]). && (31) \end{aligned}$$

The expectation of the square and the square of the expectation must remain distinct.

# 3 Convergence and useful inequalities

*Handwritten source: 9–13; L1, Sections 1.4–1.5.*

## 3.1 Convergence in probability and distribution

$$Z_n \xrightarrow{p} W \quad \Longleftrightarrow \quad \mathbb{P}(|Z_n - W| > \varepsilon) \longrightarrow 0 \quad \text{for every } \varepsilon > 0, \tag{32}$$

$$Z_n \xrightarrow{d} W \quad \Longleftrightarrow \quad F_{Z_n}(x) \longrightarrow F_W(x) \quad \text{at every continuity point of } F_W. \tag{33}$$

We also write $Z_n - W = o_p(1)$. A common target is a constant $c$. Convergence in probability implies convergence in distribution. Convergence in distribution to a constant also implies convergence in probability to that constant.

**Intuition**

Convergence in probability concerns closeness of the random values on a common probability space. Convergence in distribution concerns their laws, and does not by itself specify their joint coupling. The former describes consistency; the latter supports distributional approximations and confidence intervals.

**Example 1: independent variables.** Let $X_n \sim N(0, 1+1/n)$ be mutually independent. Their CDFs converge to that of $W \sim N(0,1)$, so $X_n \xrightarrow{d} W$. Nevertheless, the sequence cannot converge in probability to any random variable: for distinct $m, n$,

$$X_n - X_m \sim N\left(0, 2 + \frac{1}{n} + \frac{1}{m}\right). \tag{34}$$

Thus the sequence is not Cauchy in probability. In particular, for fixed $\varepsilon > 0$, the probabilities $\mathbb{P}(|X_n - X_m| > \varepsilon)$ stay bounded away from zero as distinct $m, n \to \infty$.

**Correction: Marginal distributions alone do not decide probability convergence**

The independence assumption in source p. 10 makes the first example valid. The marginal statement $X_n \sim N(0, 1 + 1/n)$ alone is insufficient, as the next coupling shows. This also clarifies the first convergence example in L1, p. 1-9.

**Example 2: a shared random component.** Let $W, Z_1, Z_2, \ldots$ be independent standard normals and define

$$X_n = W + \frac{Z_n}{\sqrt{n}}. \tag{35}$$

Again $X_n \sim N(0, 1 + 1/n)$, but now

$$\mathbb{P}(|X_n - W| > \varepsilon) \leq \frac{1}{n\varepsilon^2} \longrightarrow 0. \tag{36}$$

Hence $X_n \xrightarrow{p} W$ and $X_n \xrightarrow{d} W$. For $m \neq n$, $\mathrm{Cov}(X_m, X_n) = 1$; the sequence is correlated through $W$.

## 3.2 Continuous mapping and Slutsky's theorem

For a continuous function $f$,

$$X_n \xrightarrow{p} X \implies f(X_n) \xrightarrow{p} f(X), \tag{37}$$

$$X_n \xrightarrow{d} X \implies f(X_n) \xrightarrow{d} f(X). \tag{38}$$

If $X_n \xrightarrow{p} c$ for a constant $c$ and $Y_n \xrightarrow{d} Z$, then

$$Y_n + X_n \xrightarrow{d} Z + c, \tag{39}$$

$$X_n Y_n \xrightarrow{d} cZ, \tag{40}$$

$$Y_n / X_n \xrightarrow{d} Z/c \qquad (c \neq 0). \tag{41}$$

No independence between $X_n$ and $Y_n$ is required. The quotient may be defined arbitrarily on the event $X_n = 0$, whose probability tends to zero when $c \neq 0$.

## 3.3 Weak law of large numbers and central limit theorem

For IID $X_1, X_2, \ldots$, define $\overline{X}_n = n^{-1} \sum_{i=1}^{n} X_i$. If $\mathbb{E}|X_1| < \infty$, then

$$\overline{X}_n \xrightarrow{p} \mu = \mathbb{E}[X_1]. \tag{42}$$

If additionally $0 < \sigma^2 = \mathrm{Var}(X_1) < \infty$, the classical central limit theorem (CLT) gives

$$\frac{\sqrt{n}(\overline{X}_n - \mu)}{\sigma} \xrightarrow{d} N(0,1). \tag{43}$$

Informally, for large $n$, $\overline{X}_n$ is approximately $N(\mu, \sigma^2/n)$, equivalently $(\overline{X}_n - \mu)/\sigma$ is approximately $N(0, 1/n)$.

**Dependent observations**

Related limit theorems hold for suitable Markov chains and time series, but dependence requires additional conditions, such as ergodicity for a law of large numbers or suitable mixing and moment conditions for a CLT. A dependent-data CLT generally uses a long-run variance rather than $\mathrm{Var}(X_1)$ alone.

## 3.4 Markov, Chebyshev, and Hoeffding inequalities

For $X \geq 0$ and $\varepsilon > 0$, Markov's inequality is

$$\mathbb{P}(X > \varepsilon) \leq \frac{\mathbb{E}[X]}{\varepsilon}. \tag{44}$$

For a nonnegative error $X_n$, $\mathbb{E}[X_n] \to 0$ implies $X_n \xrightarrow{p} 0$; monotone decrease is unnecessary. This is useful for Monte Carlo algorithms and estimators. Applying Markov's inequality to $(X - \mathbb{E}[X])^2$ gives Chebyshev's inequality:

$$\mathbb{P}(|X - \mathbb{E}[X]| > \varepsilon) \leq \frac{\mathrm{Var}(X)}{\varepsilon^2}. \tag{45}$$

Thus a sequence with vanishing variance concentrates around its own mean. A fixed probability limit additionally requires the means to converge. For IID observations with $0 \leq X_i \leq 1$, Hoeffding's inequality gives

$$\mathbb{P}(|\overline{X}_n - \mu| > \varepsilon) \leq 2\exp(-2n\varepsilon^2), \qquad \mu = \mathbb{E}[X_i]. \tag{46}$$

**Correction: Assumptions for concentration (source p. 13)**

Boundedness alone does not give this Hoeffding bound for dependent observations. Independence is essential to this version; the IID assumption above matches L1, Theorem 1.6.

# 4 Conditional independence

*Handwritten source: 14; L2, Section 2.2 and Theorem 2.1.*

We write $X \perp\!\!\!\perp Y \mid Z$ when, for every $x, y$ and $P_Z$-almost every $z$,

$$\mathbb{P}(X \le x, Y \le y \mid Z = z) = \mathbb{P}(X \le x \mid Z = z)\mathbb{P}(Y \le y \mid Z = z). \tag{47}$$

When joint PDFs or PMFs exist, the following are equivalent, with identities interpreted almost everywhere on the relevant support:

$$X \perp\!\!\!\perp Y \mid Z; \tag{48}$$
$$p_{XY|Z}(x, y \mid z) = p_{X|Z}(x \mid z)p_{Y|Z}(y \mid z); \tag{49}$$
$$p_{X|YZ}(x \mid y, z) = p_{X|Z}(x \mid z); \tag{50}$$
$$p_{XYZ}(x, y, z) = \frac{p_{XZ}(x, z)p_{YZ}(y, z)}{p_Z(z)}; \tag{51}$$
$$p_{XYZ}(x, y, z) = g(x, z)h(y, z) \quad \text{for measurable nonnegative } g, h; \tag{52}$$
$$p_{X|YZ}(x \mid y, z) = w(x, z) \quad \text{for a measurable } w. \tag{53}$$

**#TODO: Derive the joint-density identity (source p. 14)**

The handwritten marker lies beside (51); its likely intent is to justify that identity. Starting from conditional independence, for $p_Z(z) > 0$,

$$p_{XYZ} = p_{XY|Z}p_Z = p_{X|Z}p_{Y|Z}p_Z \tag{54}$$
$$= \frac{p_{XZ}}{p_Z}\frac{p_{YZ}}{p_Z}p_Z = \frac{p_{XZ}p_{YZ}}{p_Z}. \tag{55}$$

Conversely, dividing by $p_Z$ recovers conditional factorization. Values where $p_Z = 0$ do not determine the conditional law.

**Supplement: Why the factorization criterion works**

If $p_{XYZ} = g(x, z)h(y, z)$, set $q(z) = \int g(x, z)\,dx$. On the support where the conditional ratio is defined,

$$p_{X|YZ}(x \mid y, z) = \frac{g(x, z)}{q(z)}, \tag{56}$$

which does not depend on $y$. Integrating this conditional density against $p_{Y|Z}(y \mid z)$ shows it equals $p_{X|Z}(x \mid z)$. Replace integrals by sums for discrete variables. This is the argument in L2, pp. 2-1–2-2.

# 5 Undirected graphical models (October 12, 2026)

*Handwritten source: 16–17; L2, Sections 2.3.1–2.3.3.*

## 5.1 Graph factorization and the global Markov property

For a finite undirected graph $G = (V, E)$, a clique is a vertex set with all distinct pairs adjacent. Let $\mathcal{C}(G)$ denote its maximal cliques. A normalized factorization is

$$p(x) = \frac{1}{Z} \prod_{C \in \mathcal{C}(G)} \psi_C(x_C), \tag{57}$$

$$Z = \int \prod_{C \in \mathcal{C}(G)} \psi_C(x_C)\, d\nu(x), \qquad 0 < Z < \infty. \tag{58}$$

Here $\nu$ is the appropriate product reference measure and the potentials are nonnegative. For strictly positive potentials,

$$\log p(x) = \sum_{C \in \mathcal{C}(G)} \log \psi_C(x_C) - \log Z. \tag{59}$$

**Correction: Log potentials and normalization (source p. 16)**

The logarithm of a product is a sum of *logarithms* of the potentials. The handwritten sum of $\psi_C$ is valid only if that symbol is redefined to mean a log potential. A normalizing constant must be included unless absorbed into a potential. Log-scale calculations help avoid numerical overflow and underflow.

The global Markov property (GMP) says that for disjoint vertex sets $A, B, S$, if every path from $A$ to $B$ meets $S$, then

$$X_A \perp\!\!\!\perp X_B \mid X_S. \tag{60}$$

Graph factorization implies GMP (L2, Theorem 2.7). A sparser graph generally asserts more conditional independences.

**One distribution, several graphs.** Suppose

$$p(x_1, x_2, x_3) = p_{12}(x_1, x_2) p_3(x_3). \tag{61}$$

It is globally Markov for both $G_1$ (only edge $1-2$, with 3 isolated) and $G_2$ (chain $1-2-3$). Indeed,

$$p(x_1, x_3 \mid x_2) = p(x_1 \mid x_2) p_3(x_3) \tag{62}$$

$$= p(x_1 \mid x_2) p(x_3 \mid x_2). \tag{63}$$

Thus $X_1 \perp\!\!\!\perp X_3 \mid X_2$, as required by the chain. It is often useful to seek a sparse graph compatible with the data-generating distribution.

## 5.2 Local Markov property and a counterexample

*Handwritten source: 18–19; L2, Definition 2.10 and Example 2.13.*

Let $N(j)$ be the neighbors of vertex $j$, and let $X_{-j}$ contain all variables except $X_j$. The local Markov property (LMP) is

$$p(x_j \mid x_{-j}) = p(x_j \mid x_{N(j)}) \quad \text{almost everywhere.} \tag{64}$$

GMP implies LMP; the converse can fail without positivity. Let $U, V$ be independent Ber$(1/2)$ variables and set

$$(X_1, X_2, X_3, X_4, X_5) = (U, U, UV, V, V). \tag{65}$$

Each of the following rows has probability $1/4$:

| $X_1$ | $X_2$ | $X_3$ | $X_4$ | $X_5$ |
|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 |
| 1 | 1 | 0 | 0 | 0 |
| 0 | 0 | 0 | 1 | 1 |
| 0 | 0 | 0 | 0 | 0 |

Consider the chain $X_1 - X_2 - X_3 - X_4 - X_5$. LMP holds: the conditional distributions of $X_1, X_2, X_4, X_5$ are determined by a neighboring duplicate, while $X_3 = X_2 X_4$ is determined by its two neighbors. Further conditioning changes none of these point masses.

**#TODO: Compute the conditional probabilities (source p. 19: "how?")**

The event $X_3 = 0$ excludes only the first row, so it has probability $3/4$. The remaining rows have conditional probability $1/3$ each. Hence

$$\mathbb{P}(X_2 = 0 \mid X_3 = 0) = \frac{1/2}{3/4} = \frac{2}{3}, \tag{66}$$

$$\mathbb{P}(X_4 = 0 \mid X_3 = 0) = \frac{2}{3}, \tag{67}$$

$$\mathbb{P}(X_2 = 0, X_4 = 0 \mid X_3 = 0) = \frac{1/4}{3/4} = \frac{1}{3} \neq \frac{4}{9}. \tag{68}$$

Thus $X_2 \not\perp\!\!\!\perp X_4 \mid X_3$, although vertex 3 separates 2 and 4. GMP fails. The handwritten $2/3$ values are marginals, not their product; the comparison needed for independence is $1/3 \neq (2/3)(2/3)$.

## 5.3 Pairwise Markov property

*Handwritten source: 20; L2, Definition 2.11 and Example 2.14.*

The pairwise Markov property (PMP) requires

$$X_i \perp\!\!\!\perp X_j \mid X_{V\setminus\{i,j\}} \quad \text{whenever } i \neq j \text{ and } \{i,j\} \notin E. \tag{69}$$

Always,

$$\text{graph factorization} \implies \text{GMP} \implies \text{LMP} \implies \text{PMP}. \tag{70}$$

To see why PMP need not imply LMP, let $X_1 = X_2 = X_3 = U$, where $U \sim \mathrm{Ber}(1/2)$, and use the graph with only edge $2 - 3$. For either missing edge, conditioning on the third variable makes both variables deterministic, so PMP holds. But $X_1$ has no neighbors and

$$\mathbb{P}(X_1 = 1 \mid X_2 = 0, X_3 = 0) = 0 \neq \mathbb{P}(X_1 = 1) = \frac{1}{2}. \tag{71}$$

Thus LMP fails. The same example works with only edge $1 - 3$ (the other graph drawn in the handwriting) or only edge $1 - 2$, by relabeling the isolated vertex.

## 5.4 Positivity and Hammersley–Clifford

*Handwritten source: 21; L2, Proposition 2.15 and Theorem 2.17.*

For discrete variables, write $\mathcal{X}_j = \{x_j : p_j(x_j) > 0\}$. Strict positivity means

$$p(x) > 0 \quad \text{for every } x \in \mathcal{X}_1 \times \cdots \times \mathcal{X}_d. \tag{72}$$

Equivalently, the positive joint support is the full Cartesian product of the marginal positive supports. Under positivity the three Markov properties are equivalent. The finite-state Hammersley–Clifford theorem further makes them equivalent to clique factorization:

$$\text{graph factorization} \iff \text{GMP} \iff \text{LMP} \iff \text{PMP}. \tag{73}$$

**Scope of the positivity statement**

For continuous variables, positivity is a statement about a density relative to a product reference measure; the topological support is generally the closure of the set where a chosen density is positive. L2 discusses continuous extensions. The finite-state theorem above is its explicit Theorem 2.17. The deterministic counterexamples violate product-support positivity.

## 5.5 Gaussian graphical models

*Handwritten source: 22; L2, Section 2.3.5.*

For a positive definite covariance matrix $\Sigma$,

$$X \sim N_d(\mu, \Sigma), \qquad p_X(x) = \frac{\exp\{-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu)\}}{(2\pi)^{d/2}\det(\Sigma)^{1/2}}, \quad x \in \mathbb{R}^d. \tag{74}$$

**#TODO: Check the Gaussian density (source p. 22)**

The handwritten normalizer uses $1/\det(\Sigma^{-1})$, which equals $\det(\Sigma)$ and is incorrect. The correct factor is $\det(\Sigma)^{-1/2} = \det(\Sigma^{-1})^{1/2}$. For example, setting $d = 1$ recovers the univariate normal factor $1/\sqrt{2\pi\sigma^2}$. A nonsingular covariance matrix is positive definite, so this density is strictly positive everywhere. A singular Gaussian does not have this full-dimensional Lebesgue density.

**Supplement: The precision matrix encodes missing edges**

Let $K = \Sigma^{-1}$. Expanding the quadratic form in $\log p_X$ gives the cross term $-K_{ij}(x_i - \mu_i)(x_j - \mu_j)$ for $i \neq j$. The conditional bivariate Gaussian given the remaining coordinates has precision equal to the principal block $K_{\{i,j\},\{i,j\}}$. Its coordinates are independent exactly when the off-diagonal entry vanishes. Thus

$$K_{ij} = 0 \quad \Longleftrightarrow \quad X_i \perp\!\!\!\perp X_j \mid X_{V\setminus\{i,j\}}. \tag{75}$$

Zeros of the covariance matrix instead encode *marginal* independence in a Gaussian model. This completes the important concept introduced in L2, Section 2.3.5.